

EDISON, THOMAS



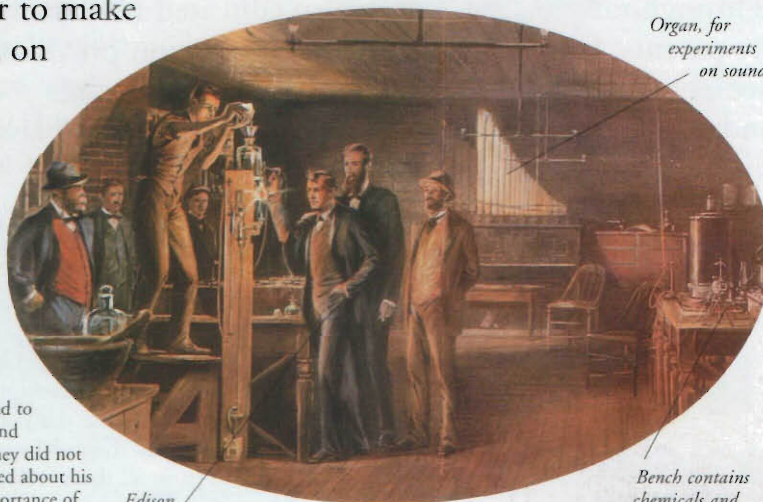
ONE OF THE GREATEST INVENTORS of all time, Thomas Alva Edison produced a number of inventions that changed the world – electric lighting, sound recording, and an early form of moving pictures, among many others. He had little formal schooling, but he was fascinated by science. He worked extremely hard, and would spend days, months, or even years experimenting in order to make something work. He often slept fully clothed on one of his worktables, so that he could start work again first thing in the morning.

Menlo Park

In 1876, using the money from his stock "ticker", Edison built an "invention factory" at Menlo Park, 39 km (24 miles) from New York City. This barn-like two-storey building was the world's first research laboratory, where a staff of scientists helped Edison to develop his ideas into devices that actually worked. In the six years that Edison worked at Menlo Park, he patented more than 400 different inventions.

Research work

At Menlo Park, Edison would come up with rough ideas and sketches. These would be refined, built, and tested by his assistants. They often had to build inventions again and again to find out why they did not work. Edison, when asked about his success, stressed the importance of these setbacks. "I failed my way to success," he said.



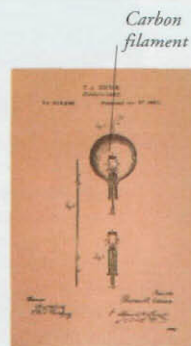
Edison watches to see how strongly the bulb glows.

Organ, for experiments on sound

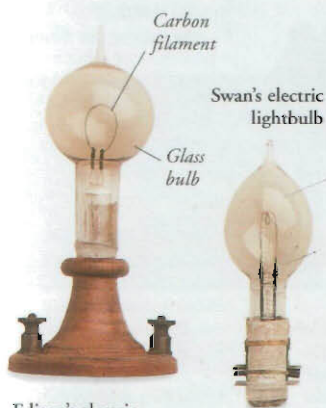
Bench contains chemicals and other scientific equipment.

Electric light

Perhaps Edison's most important invention was the electric lightbulb. He saw that a bulb with a glowing thread or filament would work, using little electricity. It took him thousands of experiments before he discovered that the best material for the filament was carbonized cotton thread. British scientist Joseph Swan (1828–1914) invented a lightbulb at the same time as Edison, and the two men later joined forces.



Patent drawing for the lightbulb



Edison's electric lightbulb

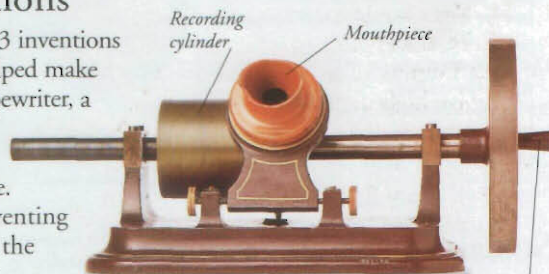


Lighting the city

Having developed the lightbulb, Edison went on to create a complete electric lighting system, powered by a central generator. His first power plant opened in 1882, serving 85 satisfied customers. Soon, whole cities were lit with electricity.

Other inventions

Edison patented 1,093 inventions in his lifetime. He helped make the first successful typewriter, a dictating machine, and an improved telephone mouthpiece. He came close to inventing radio, and predicted the use of atomic power.



Edison's phonograph

Handle to turn cylinder

Phonograph

The phonograph, a device for recording and playing back sounds, was Edison's favourite invention. He sketched the machine and gave it to an assistant to build. It worked, but Edison did not realize this because he had poor hearing.



Recording the voice

Kinetoscope

In 1889, Edison invented the kinetoscope, a projector with a peepshow-type viewer to go with it. Kinetoscopes were installed in special viewing parlours in the USA, and customers paid to watch short films.



THOMAS EDISON

- 1847 Born, Milan, Ohio, USA.
- 1869 Improves the "ticker", for relaying prices on the stock market.
- 1876 Moves to Menlo Park.
- 1877 Creates the phonograph.
- 1877 Invents the carbon microphone, for use in telephone mouthpieces.
- 1879 Patents the electric lightbulb.
- 1882 Power switched on at the Pearl Street generating station, New York.
- 1883 Edison and Swan form an electric company.
- 1889 Invents the kinetoscope.
- 1931 Dies, aged 84.

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